

Smart Workout Suit

Self-Powered Biometric Sensing System

Technical Concept & Prototype Roadmap

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Referenced research:

Flexible TEG with radiative cooling for BLE sensing — ScienceDirect 2025
Flexible TEG and energy management powered by body heat — Nature/PMC 2023

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1. Project Overview

We are building a **self-powered smart workout suit** (compression shirt + shorts) that monitors the wearer's biometrics in real time and transmits data wirelessly to a phone — with no charging required.

Core goals

- Embed multiple body sensors (ECG, IMU, SpO₂, blood pressure, temperature, humidity)
- Power everything from the wearer's own body heat and movement
- Transmit live data over Bluetooth Low Energy (BLE 5.x)
- Keep total prototype cost under **12,000**

Design principle: The suit should work during a workout without any cable or battery top-up. Energy is harvested continuously from skin heat (TEG), sunlight (flexible solar), and footsteps (piezoelectric insert).

2. Research Basis

2.1. ScienceDirect 2025 — Key Breakthrough

This is the most critical paper for our project. The researchers built an ultra-thin Flexible Thermoelectric Generator (FTEG) integrated with a **radiative cooling (RC)** layer on a soft silicone substrate — only **2.26 mm thick** total.

- Achieved a maximum normalised power density of **3.51 $\mu\text{W}/\text{cm}^2$** under realistic wearable conditions ($T_{\text{skin}} = 32^\circ\text{C}$, $T_{\text{ambient}} = 26^\circ\text{C}$).
- Stored harvested energy in a power management circuit and capacitor.
- **Successfully powered a BLE sensing module** driven entirely by body heat.

What this means for us: The TEG completely replaces the “electricity-generating fabric” problem we are stuck on. We do not need TENG fabric at all. Skin heat alone can run our BLE and sensors.

2.2. Nature / PMC 2023 — End-to-End Proof

This paper demonstrates a wearable health-monitoring bracelet where:

- The FTEG + energy management IC powers a BLE chipset, sensors, and peripheral electronics.
- The system operates at a temperature difference as low as **4 K** between skin and

ambient air.

- Real-time data is transmitted with a **cycle of just 1.6 seconds** — battery-free.
- The energy management circuit boosts the FTEG output from as low as 30.2 mV to a stable **3.4 V DC**.

What this means for us: Even in India’s warm climate (where skin-to-air ΔT is small), this system works. Our TEG just needs to be large enough — more surface area = more power harvested.

3. Power Subsystem — Roadblock Solved

The team was debating three power options. Here is the honest engineering assessment.

3.1. Option A — TENG Fabric (Rejected for prototype)

Pranay correctly identified all the problems. Triboelectric fabrics produce high voltage but near-zero current, output is intermittent (only during movement), durability drops with repeated rubbing, and washing resistance is very difficult to achieve. **Skip for the prototype.**

3.2. Option B — Flexible Solar Panel (Supplementary)

The DFRobot FIT0333 panel from Evelta (1.5 V / 250 mA, 1,201) is truly flexible and can be sewn onto the shoulder area of the suit. It is useful for outdoor workouts in sunlight, but contributes nothing indoors or at night. **Use as a supplement, not as the primary source.**

3.3. Option C — TEG Body Heat (Primary — Recommended)

This is the winner for the prototype, backed directly by the two research papers above. A TEG placed on the upper back or chest generates power continuously as long as the suit is worn.

3.4. Option D — Piezoelectric Shoe Insert (Supplementary)

Pranay’s suggestion of a piezoelectric membrane in the shoe sole is excellent as an additional source. Each footstep compresses the membrane and generates a pulse of electrical energy. This can be fed into the same power management IC as the TEG.

Power management strategy (from research): Use a boost converter to step up the TEG’s low voltage ($\sim 30\text{--}100\text{ mV}$) to 3.3–4 V. Store charge in a supercapacitor. A hysteretic switch fires the MCU and sensors only when the capacitor rises above a threshold voltage, then hibernates. This “burst and sleep” architecture is exactly how the 2023 Nature paper achieves battery-free operation.

4. Component Selection

4.1. Power Components

Component	Part Number	Source	Cost (INR)
Flexible solar panel	DFRobot FIT0333	Evelta	1,201
TEG module	TEC1-12706 (demo)	Robu / Amazon	200–500
Boost converter IC	TPS61220 / MAX17222	Mouser India	150–300
Storage supercapacitor	100–470 μ F	Amazon local	100
Backup LiPo cell	100 mAh flat cell	Robu.in	250

4.2. MCU and BLE

A single chip combining the microcontroller and BLE radio is the correct approach for minimising power consumption.

- **nRF5340** (Nordic Semiconductor) — BLE 5.3 onboard, sleeps at $\sim 1 \mu\text{A}$. Ideal for this application, directly matches the approach in the 2023 Nature paper.
- **ESP32-C3** — Cheaper, widely available in India, BLE 5.0 onboard. Good for initial prototyping.

4.3. Sensor Suite

Parameter	Chip	Notes
ECG	AD8232 module	Heart electrical signal; chest placement
SpO ₂ + pulse	MAX30102	Blood oxygen + heart rate; wrist / finger
IMU $\times$ 3	MPU-6050 / ICM-42688	Both elbows + lumbar (as in the sketch)
Skin temperature	LM35 / TMP117	Thermal load monitoring
Humidity / sweat	SHT40	Dehydration alert
Blood pressure (est.)	Derived via PPG	Indirect from wrist pulse wave
Respiratory rate	Derived from ECG/IMU	No extra sensor needed

5. Sensor Placement on the Suit

This matches the body diagram sketched by the team (back view):

- **Upper back (centre)** — ECG module + TEG patch (skin contact for power generation)
- **Left elbow + Right elbow** — IMU modules for arm motion and rep counting
- **Lumbar / lower back** — IMU for core stability and posture monitoring
- **Shoulders (left + right)** — Flexible solar panels; Ariyan's suggestion is correct
- **Wrist cuff (attached to sleeve)** — MAX30102 for SpO₂ and pulse
- **Chest (shirt front)** — ECG electrode pads (dry-contact conductive fabric)
- **Shoe sole** — Piezoelectric membrane insert for step-based energy harvesting

6. Prototype Build Plan

6.1. Phase 1 — Power Bench Test (1–2 weeks)

- Wire TEG module to boost converter; measure output voltage under skin contact
- Confirm capacitor can be charged to 3.3 V from body heat alone
- Add solar panel in parallel and verify combined output
- **Success criterion:** stable 3.3 V with no external supply while wearing

6.2. Phase 2 — Sensor Validation (1–2 weeks)

- Get AD8232 ECG working with Arduino Uno first (straightforward)
- Add MAX30102 for SpO₂ and pulse
- Add MPU-6050 IMUs at both elbow positions and lumbar (matches sketch)
- Verify all sensors communicate correctly over I²C / SPI bus

6.3. Phase 3 — Integration (2–3 weeks)

- Replace Arduino with nRF5340 or ESP32-C3
- Run all sensors from TEG + capacitor power system
- Write BLE GATT service to stream all sensor data simultaneously
- Build a simple phone app (MIT App Inventor or Flutter) for live display

6.4. Phase 4 — Embedding in Suit (2 weeks)

- Sew flat flex cables into a compression shirt and shorts
- Mount TEG on upper back with silicone pad for consistent skin contact

- Attach solar panels on shoulders with conductive thread connections
- Mount ECG electrodes on chest; IMUs at elbows and lumbar
- Insert piezo membrane into shoe sole

7. Bill of Materials — Prototype Cost Estimate

Item	Cost (INR)
Flexible solar panels × 2 (FIT0333)	2,400
TEG modules × 2	600
Power management ICs (boost converter, etc.)	500
nRF5340 development board	1,800
All sensors (ECG, SpO ₂ , IMU ×3, temp, humidity)	2,500
Backup LiPo cell	250
Wires, connectors, custom PCB	800
Compression suit base (shirt + shorts)	1,500
Piezoelectric shoe insert	400
Miscellaneous (solder, flex PCB, tape)	450
Total (estimated)	11,200

8. Showcase Strategy

We do **not** need the electricity-generating fabric for the prototype demonstration. The TEG on the back combined with solar panels on the shoulders is a complete, demonstrable, and research-backed power system.

Pitch line for showcase:

“The suit is powered entirely by body heat via thermoelectric generation — a method proven by researchers published in Nature (2023) to power BLE sensors and wireless transmission at temperature differences as small as 4°C. No battery. No charging.”

What to demonstrate live

1. Wear the suit; show live ECG waveform on phone screen
2. Show SpO₂ and heart rate updating in real time
3. Do 10 bicep curls; show IMU rep counter incrementing on screen

4. Cover the shoulder solar panels; show system continues on TEG alone
5. Show power meter: TEG generating milliwatts from body heat live

9. Suggested Division of Work

Member	Focus Area
Pranay	Overall system integration, BLE firmware (nRF5340 / ESP32-C3), phone app
Ariyan	Sensor module testing (ECG, IMU, SpO ₂), PCB layout, suit embedding
Reman Dey	TEG contact area optimisation, boost converter circuit design, power benchmarking

Reman's immediate task: Research how to maximise TEG skin contact area across the upper back while maintaining flexibility. Also study the boost converter circuit from the 2023 Nature paper (MAX17222 or BQ25570) — this is the real engineering challenge now.

10. References

1. Khan et al., “Flexible thermoelectric generator with radiative cooling for body-heat-driven self-powered Bluetooth low energy sensing system,” *Nano Energy / ScienceDirect*, 2025. DOI: [S2214993725004087](https://doi.org/10.1016/j.nanoen.2025.109937)
2. Yin et al., “Flexible thermoelectric generator and energy management electronics powered by body heat,” *Microsystems & Nanoengineering / Nature*, August 2023. PMC: [PMC10030726](https://pubmed.ncbi.nlm.nih.gov/40030726/)
3. DFRobot, “Flexible Solar Panel 1.5 V / 250 mA (FIT0333),” EVELTA Electronics. [evelta.com](https://www.evelta.com/)